Definition 1 (Deep Reasoning: $\mathcal{D}$). A transition $e_t = (u_t \to u_{t+1})$ is labeled $\mathcal{D}$ if the primary intent of u_{t+1} is to extend the reasoning chain via non-trivial inference (e.g., multi-step causal, deductive, or analogical reasoning). Operationally, $\mathcal{D}$ typically introduces at least one of: (i) new latent assumptions, (ii) intermediate logical variables/nodes, or (iii) derivations that go beyond direct computation or restatement.

Definition 2 (Self-Reflection: $\mathcal{R}$). A transition $e_t = (u_t \to u_{t+1})$ is labeled $\mathcal{R}$ if u_{t+1} explicitly comments on, audits, or regulates the model's own reasoning process. This includes expressing uncertainty, revising the solution strategy, identifying or correcting mistakes, or re-examining earlier steps (i.e., tracing back to prior logical nodes).

Definition 3 (Self-Exploration: $\mathcal{E}$). A transition $e_t = (u_t \to u_{t+1})$ is labeled $\mathcal{E}$ if u_{t+1} intentionally branches into alternative hypotheses or candidate solution paths. Operationally, $\mathcal{E}$ raises new possibilities or sub-questions and keeps multiple paths active rather than committing to a single convergent chain.

Remark (Normal Operation; $\mathcal{N}$). Any transition not meeting the criteria for $\mathcal{D}$, $\mathcal{R}$, or $\mathcal{E}$ is labeled $\mathcal{N}$. This category covers routine progression such as straightforward calculation, paraphrasing, formatting, or direct execution of an already-chosen plan.

Note: It should be noted that an edge linking logical nodes does not merely connect the step s_t to the next step s_{t+1}, but also adds edges to previously related nodes.

C.1.2 Attention Energy Definition of Reasoning Bonds.

We now replace the molecular analogy with a direct analysis of attention patterns. For a given Transformer layer and head, let $q_i, k_j \in \mathbb{R}^{d_k}$ denote the query and key vectors of the i-th and j-th tokens, respectively. The attention weight α_{ij} from token i to token j is

$$s_{ij} = \frac{q_i^\top k_j}{\sqrt{d_k}}, \tag{6}$$

$$\alpha_{ij} = \frac{\exp\left(\frac{q_i^\top k_j}{\sqrt{d_k}}\right)}{\sum_\ell \exp\left(\frac{q_i^\top k_\ell}{\sqrt{d_k}}\right)}. \tag{7}$$

Following the standard Gibbs–Boltzmann parametrization, we define the <u>attention energy</u>

$$E_{ij} \triangleq -s_{ij} = -\frac{q_i^\top k_j}{\sqrt{d_k}}. \tag{8}$$

Then, substituting Eq. (8) into Eq. (7) yields

$$\alpha_{ij} = \frac{\exp(-E_{ij})}{\sum_\ell \exp(-E_{i\ell})}, \tag{9}$$

which is a Boltzmann distribution over targets j with inverse temperature fixed to 1.[‡] That is, the attention weight is E_{ij} Gibbs–Boltzmann distribution (temperature is taken to be 1). In this view, lower E_{ij} corresponds exactly to higher attention weight α_{ij}, i.e., stronger dependency from i to j.

To relate this to Long CoT behaviors, we use the edge-level labeling procedure from Appendix C.3. Each reasoning transition (edge) in a trajectory is classified as one of three behaviors: Deep Reasoning, Self-Reflection, or Self-Exploration (plus Normal Operation as a local baseline). For each labeled edge b, we aggregate the token-level attention weights between the source and target steps into a single scalar logit, and convert it to an energy value E_b using Eq. (8). This yields three random variables $E_{\mathcal{D}}, E_{\mathcal{R}}, E_{\mathcal{E}}$ corresponding to Deep, Reflection, and Exploration edges, respectively.

[‡]We do not assume any physical semantics for E_{ij}; it is simply a convenient reparameterization of the attention logits.

C.1.3 Attention-Energy Ordering of Reasoning Bonds

To analyze ordering among attention energies for different reasoning bonds, model queries, and keys with Rotary Positional Embedding (RoPE) [35]. In an autoregressive Transformer, attention weights are a softmax of pre-softmax logits; define an attention energy as the negative logit so that higher attention corresponds to lower energy (Boltzmann/Gibbs form).

RoPE model RoPE can be written as a position-dependent block-diagonal rotation on (pairs of) coordinates. Model this head's queries/keys as

$$q_i = R(i)u_i, \qquad k_j = R(j)v_j, \tag{10}$$

where $R(t) \in \mathbb{R}^{d_k \times d_k}$ is deterministic and orthogonal. Then

$$s_{ij} = \frac{u_i^\top R(i)^\top R(j) v_j}{\sqrt{d_k}} = \frac{u_i^\top R(i-j) v_j}{\sqrt{d_k}}, \tag{11}$$

so the score depends only on the relative offset.

Bond random variables and expectations Let $d_D := 1$ (adjacent), and choose integers $1 < d_R < d_E$. Define bond-energy random variables (random over positions/data)

$$\mathcal{E}_{\mathcal{D}}(i) := E_{i,i-1}, \qquad \mathcal{E}_{\mathcal{R}}(i) := E_{i,i-d_R}, \qquad \mathcal{E}_{\mathcal{E}}(i) := E_{i,i-d_E}. \tag{12}$$

Define their mean energies (scalars)

$$\bar{E}_{\mathcal{D}} := \mathbb{E}\left(\mathcal{E}_{\mathcal{D}}(i)\right), \quad \bar{E}_{\mathcal{R}} := \mathbb{E}\left(\mathcal{E}_{\mathcal{R}}(i)\right), \quad \bar{E}_{\mathcal{E}} := \mathbb{E}\left(\mathcal{E}_{\mathcal{E}}(i)\right). \tag{13}$$

Goal: prove $\bar{E}_{\mathcal{D}} < \bar{E}_{\mathcal{R}} < \bar{E}_{\mathcal{E}}$ in expectation, and then with high probability for empirical averages.

Assumptions Here, we should establish weak dependence under RoPE and state two assumptions on the query/key distributions.

A1 (Isotropic, distance-decaying cross-covariance). There exists a scalar function $\rho(d) \geq 0$, strictly decreasing in d, such that for $d = |i - j|$,

$$\mathbb{E}[u_i v_j^\top] = \rho(d)\, I. \tag{14}$$

A2 (Positive average alignment of rotation). Define

$$\mu(d) := \frac{1}{\sqrt{d_k}} \operatorname{tr}(R(d)). \tag{15}$$

Assume $\mu(d)$ is non-increasing in d and $\mu(d) \geq \underline{\mu} > 0$ for $d \in \{1, d_R, d_E\}$.

Theorem 1 (Expected bond-energy order under RoPE). Under A1–A2 and $1 < d_R < d_E$,

$$\bar{E}_{\mathcal{D}} < \bar{E}_{\mathcal{R}} < \bar{E}_{\mathcal{E}}. \tag{16}$$

Proof. First, compute the expected logit at distance d. Let $j = i - d$. Then

$$\mathbb{E}[s_{i,i-d}] = \mathbb{E}\left[\frac{u_i^\top R(d) v_{i-d}}{\sqrt{d_k}}\right] \tag{17}$$

$$= \frac{1}{\sqrt{d_k}} \operatorname{tr}\big(R(d)\, \mathbb{E}[v_{i-d} u_i^\top]\big)\,. \tag{18}$$

By A1, $\mathbb{E}[v_{i-d} u_i^\top] = \rho(d) I$. Hence

$$\mathbb{E}[s_{i,i-d}] = \rho(d) \cdot \frac{1}{\sqrt{d_k}} \operatorname{tr}(R(d)) = \rho(d)\mu(d). \tag{19}$$

By A1, $\rho(1) > \rho(d_R) > \rho(d_E)$. By A2, $\mu(1) \geq \mu(d_R) \geq \mu(d_E) \geq \underline{\mu} > 0$. Therefore

$$\mathbb{E}[s_{i,i-1}] > \mathbb{E}[s_{i,i-d_R}] > \mathbb{E}[s_{i,i-d_E}]. \tag{20}$$

Finally, since $E_{ij} = -s_{ij}$, taking negatives yields

$$\bar{E}_{\mathcal{D}} < \bar{E}_{\mathcal{R}} < \bar{E}_{\mathcal{E}}. \tag{21}$$

The lower bound between two bond energies. Let $\widehat{s}(d)$ be the sample mean of $s_{i,i-d}$ over N independent draws (or approximately independent blocks), and define $\widehat{E}(d) := -\widehat{s}(d)$. Let $\widehat{E}_{\mathcal{D}} := \widehat{E}(1)$, $\widehat{E}_{\mathcal{R}} := \widehat{E}(d_R)$, $\widehat{E}_{\mathcal{E}} := \widehat{E}(d_E)$.

A3 (Sub-Gaussian logits). For each fixed distance $d \in \{1, d_R, d_E\}$, the centered logit $s_{i,i-d} - \mathbb{E}[s_{i,i-d}]$ is σ^2-sub-Gaussian.

Then for each d and any $\epsilon > 0$,

$$\Pr\big(\big|\widehat{s}(d) - \mathbb{E}[s(d)]\big| > \epsilon\big) \leq 2\exp\left(-\frac{N\epsilon^2}{2\sigma^2}\right). \tag{22}$$

By a union bound over the three distances, all three estimates concentrate simultaneously with probability at least $1 - \delta$ when

$$\epsilon = \sigma\sqrt{\frac{2\log(6/\delta)}{N}}. \tag{23}$$

If the expected logit gaps satisfy

$$\mathbb{E}[s(1)] - \mathbb{E}[s(d_R)] > 2\epsilon, \qquad \mathbb{E}[s(d_R)] - \mathbb{E}[s(d_E)] > 2\epsilon, \tag{24}$$

then with probability at least $1 - \delta$, that satisfies $\widehat{E}_{\mathcal{D}} < \widehat{E}_{\mathcal{R}} < \widehat{E}_{\mathcal{E}}$, since $\widehat{E}(d) = -\widehat{s}(d)$ preserves strict inequalities after flipping signs.

Lemma (Finite-sample ordering with high probability). Suppose that the mean scores satisfy $\Delta_{DR} = \mathbb{E}[s_D] - \mathbb{E}[s_R] > 0$ and $\Delta_{RE} = \mathbb{E}[s_R] - \mathbb{E}[s_E] > 0$. For any $0 < \epsilon < \min(\Delta_{DR}, \Delta_{RE})/2$, if

$$N \geq \frac{2\sigma^2}{\epsilon^2}\log\frac{4}{\delta}, \tag{25}$$

then, with probability at least $1 - \delta$, the empirical estimates follow the same ordering:

$$\Pr(\hat{E}_D < \hat{E}_R < \hat{E}_E) \geq 1 - \delta. \tag{26}$$

C.1.4 Low-energy edges dominate path aggregation.

Step-level dependency graph. Let a model-produced chain-of-thought be partitioned into reasoning steps. Represent these steps as nodes in a directed graph $G = (V, B)$. For two steps $u, v \in V$, include a directed edge $b = (u \to v) \in B$ if step v attends to step u with average attention weight above a fixed threshold. (Thus, edges point from an attended-to step to the attending step.)

Step-level edge energy. Assume token-level energies E_{ij} are defined for token pairs (i, j). For an edge $b = (u \to v)$, define the set of cross-step token pairs

$$\mathcal{C}(u \to v) \;=\; \{(i, j) : i \in \text{tokens}(u),\; j \in \text{tokens}(v)\}. \tag{27}$$

Define the step-level edge energy as the mean token-level energy across these pairs:

$$E_b \;\equiv\; E_{u\to v} \;=\; \frac{1}{|\mathcal{C}(u \to v)|}\sum_{(i,j)\in\mathcal{C}(u\to v)} E_{ij}. \tag{28}$$